

SWAPNANIL BALA

AI Engineer · Machine Learning & Full-Stack Engineering

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EDUCATION

Northeastern University, Houry College of Computer Sciences

M.S. in Data Science

Boston, MA

Sep 2025 – Dec 2027 (Expected)

Maulana Abul Kalam Azad University of Technology

B.Tech in Electronics and Communication Engineering

West Bengal, India

Aug 2019 – Aug 2023

TECHNICAL SKILLS

Languages	TypeScript, Python, JavaScript, SQL
AI & LLM Eng.	OpenAI GPT-4o Vision, Anthropic Claude API, MediaPipe, multimodal inference, prompt design, structured response parsing
Machine Learning	PyTorch, Scikit-Learn, XGBoost, transformer fine-tuning (DistilBERT, RoBERTa), NLP, SHAP explainability
Data & Analytics	Pandas, NumPy, EDA, hypothesis testing (t-test, chi-square, ANOVA), Matplotlib, Seaborn
Frontend	React, Next.js, PWAs (service workers, web workers), i18n, Lighthouse performance optimization
Backend & APIs	Node.js, REST API design, edge functions, OAuth 2.0, Postgres Row-Level Security, rate limiting, LRU/TTL caching
Data & DevOps	PostgreSQL (NeonDB, Supabase), Zod schema validation, Vitest/Jest, Git, Docker, Vercel

PROJECTS

Lagna Atelier — Local-First Vedic Astrology Engine

Web (PWA) | Deployed

TypeScript · Next.js · astronomy-engine · Zod · GPT-4o Vision · MediaPipe · Vercel

- Built a local-first PWA (~63K lines) that runs with no backend server, computing every chart client-side and persisting user data in localStorage — zero server round-trips in the core product.
- Implemented a sidereal ephemeris layer over astronomy-engine with IAU-2006 precession, supporting 6 ayanamsha models and 6 house systems.
- Engineered 15+ derived calculation engines covering 23 divisional charts, Vimshottari dasha periods, and 51 yoga detections.
- Designed a Zod-validated rules-as-data DSL for yoga definitions, with a Monte Carlo harness measuring each rule's empirical rarity across simulated chart populations.
- Delivered a multimodal palm-reading feature on GPT-4o Vision with MediaPipe hand landmarking, covering image ingestion, prompt design, and structured response parsing behind rate-limited edge functions.
- Optimized delivery with service workers, web workers for heavy computation, and LRU/TTL caching; localized to 6 languages; backed by 561 tests.

Robust Health — AI Fitness & Wellness Platform

Full-Stack Web | Deployed

TypeScript · Next.js · Anthropic API · NeonDB (Postgres) · Supabase Auth · Vercel

- Built and deployed a multi-role web application with member and trainer dashboards, tiered subscriptions, and CSRF protection.
- Served personalized, AI-generated workout plans through the Anthropic API via Next.js API routes backed by NeonDB.
- Implemented authentication with Supabase Row-Level Security and OAuth 2.0 providers (Google, GitHub, Discord).
- Cut Lighthouse-measured load time ~50% on web and ~67% on mobile through next/image compression and route-level code splitting.

Fake News Classifier — Transformer Fine-Tuning & Explainability

Applied NLP Research

Python · PyTorch · DistilBERT · RoBERTa · SHAP · Scikit-Learn

- Fine-tuned DistilBERT and RoBERTa on the LIAR dataset for multi-class misinformation detection.
- Applied SHAP for token-level explainability, surfacing which phrases drove each classification.
- Benchmarked both transformers against a TF-IDF and logistic regression baseline, delivering a structured precision, recall, and F1 comparison across all three approaches.

EXPERIENCE

P2G Mobility Tech Private Limited (Pointo) — EV-Financing Startup

Kolkata, India

Program Analyst Intern, Technology

Nov 2024 – Jan 2025

- Consolidated and cleaned operational datasets feeding internal dashboards, resolving missing values, recovering lost source files, and standardizing records for weekly reporting.
- Wrote SQL against the production database to build recurring operational reports for the project lead.
- Supported customer acquisition in the field, presenting the pay-to-own financing program to e-rickshaw operators and framing the lithium-ion upgrade against legacy lead-acid batteries.

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate — TikTok claims classification: NLP pipeline flagging policy-violating content at ~99.5% recall (5 misclassifications across 3,817 test cases). Waze churn prediction: EDA and hypothesis testing (t-test, chi-square, ANOVA); tuned logistic regression, Random Forest, and XGBoost with GridSearchCV.